CHAPTER 12

TRANSFER OF THERMAL ENERGY

Key idea: heat flows from hot to cold by conduction, convection and radiation.

1 THREE MODES

- **Conduction:** through a material without bulk movement, by lattice vibrations and, in metals, free electrons
- **Convection:** bulk movement of a fluid caused by density differences
- **Radiation:** electromagnetic waves (mainly infrared); no medium needed

2 CONDUCTION

$$\frac{Q}{t} = -kA\frac{d\theta}{dx} = \frac{kA(\theta_1 - \theta_2)}{L}$$

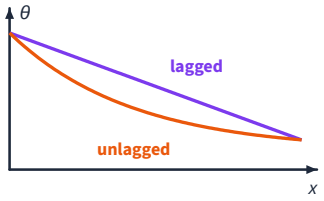
- k: thermal conductivity, W m^{-1} K^{-1}
- $d\theta/dx$: temperature gradient, K m^{-1}
- Minus sign: heat flows down the gradient

3 THERMAL RESISTANCE

$$R = \frac{L}{kA} \qquad \frac{Q}{t} = \frac{\Delta\theta}{R}$$

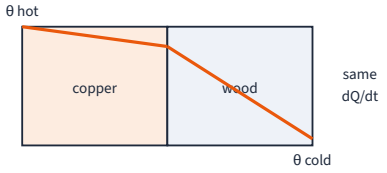
- Layers in series: $R = R_1 + R_2 + \ldots$
- Steady state: **same** rate of heat flow through every layer
- Unit K W^{-1}

4 LAGGED & UNLAGGED BARS



- Lagged: no heat lost from the sides, uniform gradient (straight line)
- Unlagged: heat lost from the sides, gradient decreases along the bar

5 COMPOSITE WALL



$$\frac{k_1 A(\theta_1 - \theta)}{L_1} = \frac{k_2 A(\theta - \theta_2)}{L_2}$$

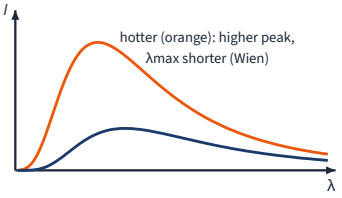
Solve for the interface temperature θ first.

6 RADIATION

$$P = e\sigma AT^4$$

$$P_{\text{net}} = e\sigma A(T^4 - T_0^4)$$

- $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}
- Black body: perfect absorber and emitter, $e = 1$
- T **must** be in kelvin



7 STARS AND PLANETS

$$A = 4\pi r^2 \qquad I = \frac{P}{4\pi d^2}$$

- Intensity at distance d from a source of power P
- Equilibrium: power absorbed = power emitted

8 COMMON SECTION A TRAPS

- Using °C in T^4; a temperature **difference** may be in °C or K.
- Using total length instead of each layer thickness.
- Different rates in series layers: in steady state they are equal.
- Saying radiation needs a medium.

EXAM TIPS

- ✓Draw the temperature profile through a composite wall.
- ✓The layer with the smaller k has the steeper gradient.
- ✓Use ratios for $P \propto T^4$: doubling T gives 16 times the power.
- ✓Quote k with unit W m^{-1} K^{-1}.